COVTL Lab Console — Technical Reference

Bridge API, anchor instrumentation, clocks, caching, extension guide

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1 Architecture

No offline mode. The browser demo engine was removed: the console has no offline mode. Without the bridge every action stays disabled behind an explicit offline banner, and the probe

runs over candidate bridge origins (same origin first, then 127.0.0.1:8421), engaging automatically when the bridge answers. The single Synthesize button speaks the blue-framed source: main input (deterministic G2P), edited final SAMPA, or compiler blocks — one precedence rule, one button.

Three tiers, all local:

```
browser UI (ES modules, no build step)
  |  fetch JSON over http, 30 s timeout, no-cache
FastAPI bridge  (server.py, 127.0.0.1:8421)
  |  in-process import, one Pipeline per job on a thread
covtl-pipeline  (vtl_synth)  ->  VocalTractLab 2.4 natives
```

The Python engine is the single source of truth for every phonetic decision; the browser only previews, displays and controls. When `vtl_synth` is not importable, the bridge answers `/api/status` with `backend:"mock"` and every engine-dependent endpoint returns a structured 503; the UI stays in an explicit offline state (banner naming the start command, buttons disabled) and its probe keeps retrying candidate bridge origins until the bridge answers — there is no simulated engine.

1.1 Module map (`covtl_console/ui/js/`)

Since v3.1 the console is a pip package: the bridge implementation lives in `covtl_console/server.py` (the UI assets it serves in `covtl_console/ui/`; a repo-root `server.py` shim keeps `python server.py` and `import server` working), and the `covtl-console` entry point (`--host`/`--port`) starts it from a plain `pip install` — see the flash install in the README. Module map:

`data.js`	Engine constants mirrored client-side: inventory, suggestion table, per-language examples, keyboard map.
`render.js`	DOM renderers: token strips, guard banner, progress, result summary, downloads, examples, history; `$`, `$$`, `el`, `toast`, `log`.
`api.js`	Transport (`jfetch` with 30 s abort, multi-origin probe), the `api` facade, client-side caches (cleared when the bridge re-engages), `normalizePolar`, `getApiBase`.
`polar.js`	`PolarStage`: the continuous display — layered canvases, dual clocks, anchor stepping. Drawing policy (journey-centric): exactly ONE journey is displayed — the anchor pair under the head. Its vocalic branch is a single arc from start anchor to end anchor (full opacity); its consonantal excursion (closed loop dep → C targets → arr) is drawn jointly when the segment carries finite c samples. Everything else is eliminated — no watermark of past journeys; the tuilage is a clean switch at the anchor boundary. Reduced-vowel anchors (`@`, `6`) are hollow rings; phonetic labels cover the current journey only. The journey's origin anchor is a red dot and its target anchor a green ring, drawn above the head (stacking visibly on stationary holds), superseding the former amber "next anchor" ring.
`spectrogram.js`	`SpectrogramView`: FFT spectrogram; `setSource` accepts the anchor list for the overlay.
`compiler.js`	`SyllableCompiler` panel; tile labels read the server-side `notation` field (single definition).

`app.js`	Controller: state machine, flows (preview / compile / synthesize / playback), transport wiring, keyboard, settings (`covtl.settings.v3`), history (`covtl.history.v1`).

2 The bridge API

All bodies are JSON. Errors use one structured envelope everywhere (v1-compatible `detail` plus a v2 error object):

```
{ "detail": "<message>",
  "error": { "class": "SampaGuardError", "message": "...",
             "hint": "...", "symbols": [ {symbol, position,
                                          reason, suggestions} ] } }
```

2.1 Endpoints

GET /api/status	Identity + caches: `backend` (`live|mock`), `engine:"vtl-synth"`, `version`, `bridge:"3.0.0"`, speakers, languages, defaults, cache counters (preview/compile/synth entries + hits), jobs, inventory digest, `out_root` (absolute). When live: `active` {speaker, lang} and lexicon sizes.
GET /api/health	`status` in `ok|degraded|offline`, `engine_version`, `active_speaker`, `active_lang`, `registry_warnings` (speaker probe failures, absent lexicons), `sandhi` capability block (below).
GET /api/languages	Six entries {`id, name, prosody_profile, lexicon_present`} + the active language.
GET /api/speakers?lang=	Registry {`id, label, f0_default, verified_with_lang, active`}; `warning_when_unverified` is set for non-English languages (vowels calibrated on `jd3` only); `calibration_speaker` is `jd3`.
POST /api/preview	Request {`text?, sampa?, mode: ortho|phonetic, lang, expressive, policy?, sandhi: off|minimal|standard|careful, sandhi_mode: g2p|post|pre`}. One round trip for the whole preview stack: response {`g2p_string, enriched_string, final_sampa, words[], groups[], warnings[], tokens[], rejected_symbols[], language, lexicon, inventory_digest, cache`}. Passing `sampa` re-enriches an edited G2P string; phonetic mode runs the guard only. A non-`off` `sandhi` level is applied through the *same gateways* as `Pipeline.run` (below) and degrades to `off` with a warning when Branch B is missing, the language has no rule module, or the input is a phonetic string; a bad level/mode is a 422. LRU(128); repeats answer `cache:true`.
POST /api/preview/sampa	Guard-only tokenization: {`tokens, rejected`}. (Deprecated v1 `/api/preview/g2p` still works, forwarding to the unified preview.)

POST /api/compile	Request {text, lang, timings, guard_policy, expressive, sandhi, sandhi_mode}. Runs the engine's anchor walk with *no synthesis*: response carries anchors[] (each anchor exports mute / syl_onset; muted anchors are labeled with the vowel their point derives from plus *), blocks[], nodes[], metrics{anchors, segments, blocks, syllables, clusters, consonants, vowels, steps, duration_s}, segments_preview[] (per anchor pair: label, notation = from·consonants·to — the tile grammar, defined once in segment_notation() and shared with /api/job/{id}/polar — cons/vowel, kinds, n_steps), per-phoneme targets{rho, theta, category}, rejected symbols, warnings, the sandhi capability block. A requested sandhi level is a documented no-op here (SAMPA blocks carry no orthography — the level acts at the text→SAMPA stage) and yields a warning. Guard error + rejected ⇒ 422 SampaGuardError. LRU(64).
POST /api/synthesize	Request (superset): {text, mode, lang, speaker, expressive, f0, timings, engine, ortho, outputMode: audio|sagittal|polar, guardPolicy, outputDir, label, exports, sandhi, sandhi_mode}. Guard error is validated *before* any job starts (⇒ 422). Response {job_id, cached, echo}; echo repeats the full resolved parameter set (the log line), including sandhi/sandhi_mode and a sandhi_note degradation notice when Branch B is missing. The level is passed to the Pipeline when its signature carries it (signature-guarded like every engine option); the synth cache key includes it. Cache hit requires state done + all artifacts on disk.
GET /api/job/{id}	Job payload: state (queued|running|cancelling|done|error|cancelled), stage (g2p|trajectory|vtl|wav|video), progress, polar_ready, wav_ready, anchors[], block_count, duration_s, echo, result{sampa, warnings, duration, frames, phrases, files[], wav_url}, error fields.
GET /api/job/{id}/polar	Segment payload (below).
POST /api/job/{id}/cancel	queued jobs become cancelled immediately; running jobs flip to cancelling (result discarded at the next checkpoint).
GET /api/job/{id}/files/{name}	FileResponse for any artifact registered on the job.

2.2 Cross-word sandhi (covtl-sandhi)

The bridge integrates the engine's *Branch-A hook* (vtl_synth.utils.sandhi, default off) without ever re-interpreting it. Architecture and rules live in the companion covtl-sandhi package (*Branch B*, six languages: fr liaison, en a→an + onset linking, it raddoppiamento + euphonic d, es enlace + y→e/o→u, pt encadeamento + EP voicing, de Glottalversatz), installed on the engine either by copying its sandhi/ folder over the engine's stub or by pip.

- **Capability probe.** At import the bridge resolves the rule package exactly as Pipeline does (stub, then top-level import sandhi); until SANDHI_FULL is true the capability stays

unavailable and `sandhi_capability()` reports {`available: false, reason: ...`}. No consumer can crash the bridge on a missing package: the preview degrades with a warning, the job echo carries a `sandhi_note`. With the package installed (`pip install covtl-sandhi`) the capability turns *available* at the next bridge start and lists the six rule languages.

- **Levels** `off|minimal|standard|careful` — a monotone coverage filter over each language's rule set (minimal ⊆ standard ⊆ careful; it decides which word-pair classes may fire, never the phonology of a firing rule). **Modes** `g2p` (liaison-aware per-word gateway + pair merge, recommended), `post` (SAMPA post-processor), `pre` (orthographic rewrite before the g2p).
- **Defaults.** The engine and every request model default to `off` (the engine CLI default — API callers get the bit-identical baseline); only the UI preselects `standard` (state `sandboxhi`, persisted in settings, sent explicitly on every preview/synthesis call).
- **Preview path.** `/api/preview` mirrors `Pipeline.run`'s degradation order (off for phonetic input, off without Branch B, off without a rule module for the language — each with a printed warning) and then applies the same three gateways; the expressive path wraps the per-word g2p in `sandhi_oracle` and closes with `apply_sandhi_sampa` (never across a `%` breath pause). The word chips use the same per-word oracle.
- **Synthesis path.** `/api/synthesize` passes `sandhi`/`sandhi_mode` to the `Pipeline` when its signature carries them; `_resolve_sampa` applies the same gateways so the early polar stage compiles the SAMPA the engine will actually speak (compile plan == synth plan). The cache keys (preview, compile, synth) all include the level.
- **No per-stretch link level.** The engine's articulation plan has no `link_level` attribute: a liaison consonant inserted at the text→SAMPA gateway appears in the compiled plan like any onset cluster, and the polar display follows the engine geometry unchanged (WYSIWYG contract). The nasal hiatus labels (`a .a`, `a.a` , merged `V˜.V˜ → a .a`) are the separate V1V2 feature of the plan itself.

2.3 The segment payload

```
{ job_id, segment_count, from, rho_max, duration_s, sample_rate: 100,
  anchors:  [ { i, t, kind, label, long, breath, hold, word_end, pt } ],
  segments: [ { segment_index, t0, t1, anchor_from, anchor_to,
                label_from, label_to, blocks[...], markers{...},
                points: [ { t, rho_v, theta_v, rho_c, theta_c } ] } ] }
```

Points run at 100 Hz; null channels mark an inactive branch (the consonantal branch is null outside clusters). `theta` arrives pre-converted to the display convention and `rho` normalized over the job's radius range, so the browser never re-derives engine units.

3 Anchor-timing instrumentation

3.1 Instrumented replay

The bridge never re-implements the walk. Inside `vtl_synth.core.syltraj.build_phrase_pval`, the engine's `build_global_pval` iterates over consecutive anchor pairs; every appended block constructs a `SylBlock`, and every iteration calls `_collect_consonants` exactly once (after the

possible departure plateau). `server._instrumented_phrase_build` hooks *both* — the same monkeypatch idiom as `vtl_synth.video.polar_video._recorded_build_pval`:

- a `RecBlock` subclass records construction order into a ledger;
- a wrapped `_collect_consonants` records the ledger length at each anchor-pair iteration.

Anchor k's step position is the cumulative block step count at iteration k, minus that iteration's departure plateau (a plateau at the slice tail belongs to the *next* iteration — iterations never emit plateaus after their hook). The terminal anchor is the total step count.

3.2 Drift checks

Two hard invariants raise `StructureDrift` (HTTP 500) instead of guessing: recorded blocks == engine blocks, and iteration count == `len(anchors)`-1. If the engine's walk ever changes shape, the bridge fails loudly.

3.3 Start-early and the `.polar` mapping

The synthesis worker builds the trajectory structure *first* (cheap, pure Python): `structure_compile` for anchors/blocks, then the engine's own `build_polar`/`write_polar` with the same parameters the engine will use — the `.polar` content is byte-identical to what `Pipeline.run(polar=True)` writes. `polar_ready` flips *before* the VTL stage starts; if the early stage fails, the display waits for the engine's own export and the job carries a note.

`.polar` format v4 is JSON at 100 Hz: `trajectory.vocalic|consonantal.{rho, theta}` in *trigonometric radians* (0 East, CCW), `phonemes[]` with t_0/t_1 spans and category, `inventory[]`. The bridge converts for display when serving the segment payload:

```
theta_display = (90 - degrees(theta_trig)) mod 360   # 0 = top, clockwise
rho_norm      = rho / rho_max ,  rho_max = max(1.45, inventory rhos)
```

4 Clock model

- **Polar-only mode.** From `polar_ready` until the audio is attached, `PolarStage` advances on its own monotonic clock (performance timestamps), scaled by the speed selector.
- **Audio mastery.** Once the WAV is ready and playing, the audio element is master: the loop reads `audio.currentTime` (read-only) each tick. The handoff at block boundaries goes through `_pendingAudioSync` so the head never jumps.
- **Autoplay fallback.** If the browser refuses autoplay, the own clock keeps the display alive; the audio takes over on its `play` event (the first user gesture on *Play* unlocks it).
- **Replay.** When the audio runs to its end, the display replays from 0 on its own clock.
- **Background-tab watchdog.** `requestAnimationFrame` freezes in hidden tabs; a 50 ms interval watchdog advances the loop whenever rAF stalls (>120 ms), so start-early works even when the console tab is hidden. Rendering resumes on focus.

5 Caching

	Key	Capacity / notes
Preview LRU	(text, sampa, mode, lang, expressive)	128; hits answer `cache:true`
Compile LRU	(text, lang, timings JSON)	64; same `cache` flag
Synthesis	SHA-1 over the resolved parameter set *including the engine version*	32; hit requires `done` + artifacts on disk

The synthesis key *excludes* `guardPolicy` deliberately: `warn`/`off` yield identical audio, and the `error` policy is re-validated on every request (fresh or cached) so a cached job can never bypass the guard. Including the engine version invalidates entries across engine upgrades instead of serving old-engine audio.

Client side, `api.js` keeps mirrors for instant UI replay; `clearCaches()` wipes them when the probe flips

6 Front-end state machine and WYSIWYG paths

`empty`	start / after clearing input.
`preview-ready`	a preview with no rejected symbols.
`guard-warn`	rejected symbols under policy `warn`; synthesis relabeled "Synthesize anyway (N dropped)".
`guard-error`	rejected symbols under policy `error`; synthesis blocked.
`synthesizing`	job running (stage strip + progress bar).
`polar-running`	start-early: polar display alive while the WAV renders.
`ready`	job done, artifacts available.
`playing`	playback in progress.

A v3 rule: *late preview responses can never downgrade a running job* — `setPreviewState` only transitions from preview states (`empty`, `preview-ready`, `guard-*`, `backend-offline`). Stale responses are dropped by a sequence counter (`previewSeq`); the compiler has the same guard for in-flight compiles.

Input reset (`resetInputs`, key `R`). One action clears both source panes and the whole preview chain: drafts, `preview`, `editedSampa`, `sampaTokens`, `sampaRejected`, plus the editor widget and guard banner. It also clears every surface built on that chain: the compiler panel (`compiler.reset()` — field, tiles, chips, cards, notes, sequence bump against in-flight compiles) and the acoustic synthesis results (playback stopped, audio element detached, polar payload dropped, spectrogram cleared, scrub ticks, result summary, downloads menu; a running job is cancelled). It also bumps `previewSeq` (an in-flight preview response is dropped by the same stale-guard) and cancels the debounced auto-preview timer, so no late response can repopulate the cleared stack. The transition to `empty` is forced with `setAppState`: the no-downgrade rule in `setPreviewState` protects against *late preview responses*, not explicit user actions. Settings persist (minus the cleared sources); language, speaker, timings and history are untouched. This exists because stale chain state leaked across input modes — an *edited* SAMPA kept steering synthesis after a pane switch, and a compiled plan stayed on screen after the input was cleared (observed).

WYSIWYG synthesis. Three paths, one promise — the string you see is the string you hear:

1. *deterministic-ortho*: **S** uses the previewed final SAMPA (the same G2P callable the engine's `Pipeline.run` uses);

2. *phonetic / edited*: **E** (or phonetic mode) sends exactly the edited string (`use_g2p=False` semantics);

3. *compiler-edited*: "Use in phonetic input" pushes the compiled string to the stage, previewed, then synthesized.

7 The single-file bundler

`build_standalone.py` inlines the ES modules into `covtl-console-standalone.html` with a purely textual transform: each module becomes an IIFE factory registered in a `__mods` registry; `import {a, b as c} from './m.js'` becomes `const {a, b: c} = __mods.m`; `export const/function/ class` are rewritten to plain declarations and the factory returns the exported names. Order is fixed (data first, app last).

Two caveats worth knowing:

- the export-name scanner accepts `$` in identifiers (regex `[A-Za-z_$][\w$]*`) — names like `$` and `$$` from `render.js` must survive the rewrite or the bundle breaks at runtime (this bit v2; fixed);

- the injected preamble installs `window.onerror` and an `unhandledrejection` hook that write the failure into `document.title`, because `file://` pages show no console.

The build is rebuildable at any time (`python build_standalone.py`); `tests/` verify the bundle compiles (`node --check`), registers all 8 modules, keeps no ES-module imports and carries the per-language example labels.

8 Test surfaces

Four lanes, all driven from `tests/`:

- **Fast suite** (`pytest tests/ -m "not slow"`): API contract, guard semantics, packaging, structure — TestClient over the in-process bridge; engine-required tests skip with an explicit reason when `vtl_synth` is absent.

- **Slow lane** (`-m slow`, nightly CI): real synthesis jobs, the MP4 path.

- **Flash e2e** (`tests/flash_e2e.py`, CI job `flash`): builds the wheel, installs it into a scratch venv (engine and covtl-sandhi resolve from GitHub through the wheel metadata), starts the installed `covtl-console` entry point and replays the live chain over HTTP — health, fr preview with liaison, compile metrics, start-early ordering, WAV-versus-structure duration, and the nasal-hiatus VO gate measured inside the engine's own `nasal_hiatus_intervals` windows (tract column 7, $\geq$0.4 max / 0.3 middle-half mean; oral control $\leq$0.2) — then uninstalls and deletes the venv. The VO probe (`_flash_vo_probe.py`) runs inside the scratch interpreter, where the engine lives.

- **Frontend checklist** (`test_frontend_checklist.md`): the manual browser series for release sign-off.

9 Known environment quirks

- **Embedded-browser loopback latency.** Requests from a webview-style browser to 127.0.0.1 can take seconds even when `curl` answers in milliseconds. The client timeout is therefore 30 000 ms (`api.js`), and every button relabels to its in-flight state so slow requests are never silent.
- **Server warm-up is synchronous.** The first G2P call loads lexicons and rules (seconds on a cold start). `__main__` runs the warm-up *before* uvicorn starts; a thread-at-import version deadlocked uvicorn's event loop on per-module import locks (observed), so the `startup` event variant only covers `uvicorn server:app`, where imports are complete when the loop starts.
- **No-cache asset policy.** `/`, `*.js`, `*.css`, `*.html` are served with `Cache-Control: no-cache` — a stale cached `app.js` once hid new features entirely.
- **Port discipline.** The bridge binds `127.0.0.1:8421` only; the v2 workbench's 8420 is untouched and both can run side by side.
- **Shared origin state.** Settings live in `localStorage` (`covtl.settings.v3`); two consoles on the same origin and port share them — expected when v3 replaces v2 at the same port, surprising when testing both at once.

10 Extension guide

Add a language example. Append to `EXAMPLES` in `js/data.js`: {`label, text, mode, lang`}. The chip row, the standalone bundle and the tests' bundle check all derive from it (rebuild the standalone). Nothing server-side changes.

Add a panel. Follow the compiler pattern: a root element in `index.html` (or one your module creates), a class in its own `js/` file exporting the panel, wiring in `app.js`, CSS tokens from `css/app.css`. Add the module to `build_standalone.py`'s `ORDER` (dependency order, data first, app last).

Add an endpoint. Extend `server.py`: a Pydantic request model, the route, and structured errors via `api_error`/`guard_error`. Keep the v2 envelope. If the endpoint produces artifacts, register them in `job["paths"]` so `/api/job/{id}/files/{name}` serves them, and respect the `exports` flag for what is *offered* versus kept for the display.

The acceptance rule. Whatever you add, the brief's acceptance rule binds the display to the engine's answer: *block boundaries shown = the anchor list returned by the engine.* Never re-segment in the browser; if the engine's walk changes shape, let `StructureDrift` fail loudly rather than guessing.